STA 35C: Statistical Data Science III

Lecture 1: Introduction and Overview

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Agenda

- Course overview
- Course logistics
- Intro to probability

What is statistical data science?

- **Statistics**: The study of collecting, analyzing, and drawing conclusions from data
- **Data science**:
  - Interdisciplinary: statistical thinking + programming + databases
  - Emphasize real-world data wrangling, practical computing, and applications (science, engineering, business, sports, government, etc.)
- **STA 35 sequence**: Introductory data science courses from a statistical perspective, with an emphasis on computing with R
  - **STA 35A**: Introduction to statistics
    (probability, distributions, confidence intervals, hypothesis testing ,etc.)
  - **STA 35B**: Advanced R functionalities + additional statistical methods
    (linear regression, ANOVA, permutation tests, etc.)
  - **STA 35C**: Fundamentals of **statistical learning** methods
    - understanding key ideas, how and when to apply them, and their limitations

What is statistical learning?

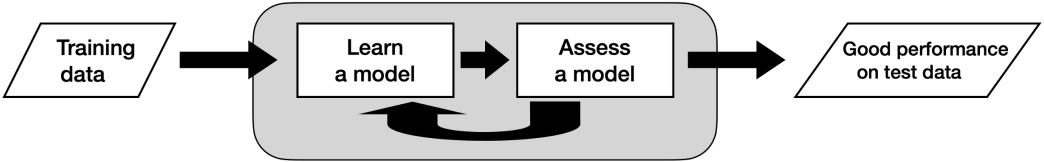
- **Definition**: A set of tools for understanding data and making informed predictions
- **Examples**:
  - Identifying critical disease risk factors from large patient records
  - Predicting whether an event will occur (e.g., credit default, septic shock)
  - Classifying medical images or tissue cells (e.g., benign vs. malignant tumors)
  - Recognizing and localizing objects in images (e.g., for autonomous vehicles)
  - Evaluating impacts of new legislation or predicting unemployment rates
  - Modeling relationships among many variables to gain practical insights (e.g., which marketing strategies drive sales)
  - ...

Supervised vs. unsupervised learning

- **Supervised learning**
  - **Setup**: We have measurements of an outcome Y and predictors (features) X
  - **Goals**: Predict Y (given X), or understand which X affects Y and how
  - **Examples**:
    - *Regression*: Forecast a product's sales next month
    - *Classification*: Predict if a customer will default on a loan

- **Unsupervised learning**
  - **Setup**: We only have features X, without any outcome variable
  - **Goals**: Discover hidden patterns or groupings in the data
  - **Examples**:
    - *Dimension reduction*: Extract or combine a few features for compression
    - *Clustering*: Cluster customers by purchasing behavior

Statistical learning and STA 35C



- **Core idea**: Learn a model from training data, evaluate its performance, and refine it
  - Aim for good predictions or insights on ***new, unseen* data**
  - Rely on probability and statistical principles to handle uncertainty
- In **STA 35C**, you'll learn the fundamentals of these methods
  - When and how to use different supervised or unsupervised learning methods
  - How to assess and interpret models (cross-validation, model selection, ...)
- Our focus is on learning **first principles**; we **do not** cover advanced machine learning techniques (e.g., deep neural networks, large language models)

Course logistics

- **Instructor:** Dogyoon Song
  - Email: dgsong@ucdavis.edu
  - Office hours: Wednesdays, 3:30–4:30 PM (or by appointment) at MSB 4220
- **TA:** Yanhao Jin
  - Email: yahjin@ucdavis.edu
  - Office hours: Tuesdays, 12:30–2:30 PM at MSB 1143
- **Lectures:** Monday, Wednesday and Friday, 1:10–2:00 PM at Wellman Hall 6
- **Lab/Discussions:** Tuesday 8:00–8:50 AM / 9:00–9:50 AM at TLC 2212 (run by TA)
- **Online platforms**
  - Course webpage: Lecture notes, homework, supplementary materials, etc.
  - Canvas: Homework submission, solutions and grades, lab materials
  - Piazza: Announcements and discussion
  - Email: Private matters only (**do not** send messages on Canvas)

Course content & prerequisites

Course content:

- Probability basics
- Intro to supervised learning
  - Fundamental concepts
  - Regression
  - Classification
- Model assessment, selection, and inference
  - Cross-validation and the bootstrap
  - Model selection and regularization
  - Simultaneous inference
- Intro to unsupervised learning
  - Dimension reduction and PCA
  - Clustering

Prerequisite(s):

- STA 035B (C- or better)
- MAT 016B or 017B or 021B (C- or better)

These requirements are strict

- If you don't meet prerequisites, please submit a petition ASAP
- Try "Homework 0" for self-assessment

“Homework 0” for self-assessment

- Complete the “Homework 0” for your self-assessment ASAP if you haven’t yet
- It reviews key material from STA 35A/B, and briefly check on your familiarity with R
- It will not be collected or graded, and no solutions will be provided
- If you struggle with any part, review your notes/resources, **attend discussion sections tomorrow** (Tue, March 31, 2026) or bring questions to office hours

Grading

- **Homework**: 35%
  - 7 homework assignments, excluding "Homework 0"
  - Assigned on Wednesday morning, due next Tuesday 11:59 pm PT
  - One homework with the lowest score can be dropped
  - **No late homework accepted for any reason**
- **Midterm exams**: 25%
  - Two in-class midterms (Fri, April 24 & Fri, May 15)
  - The lower can be dropped
  - **No make-up exams offered**
- **Final exam** (cumulative): 40%
  - Friday, June 5, 1:00–3:00 PM
- **Participation**: up to 3% extra through in-person participation in lecture/discussion

See syllabus for full details and additional information (textbook, course policies, etc.)

Software: R and RStudio

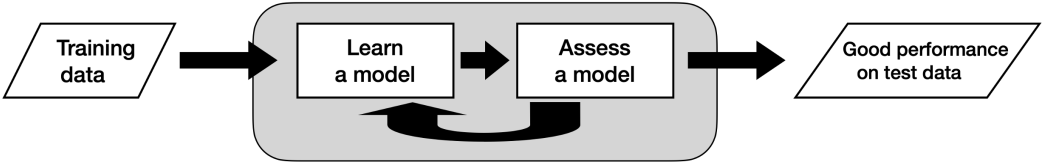
- **R** is a free, open-source **statistical programming language** for data analysis:
  - Interactive environment for data wrangling, modeling, and visualization
  - Highly extensible via packages
  - In this course, you will use R for homework and labs
- **Where to run R / Computer access**:
  - You can install R and RStudio on your own computer
  - Alternatively, refer to UC Davis Cloud Services
  - If there is a problem, let us know immediately—resources are available to help
- **Lab/discussion section**:
  - Held in TLC 2212, where computers are available
  - You may also bring your own laptop (please charge it beforehand)
  - Lab/discussion and TA office hours are the best places to get help with R

Motivation: Why probability?

Recall the goals of statistical learning:

- Predict Y given X (learn a function $f : X \to Y$)
- Identify patterns in data X

Standard workflow:



Key challenge:

- We aim for good predictions or insights on ***new, unseen* data**
- How should we assess a model, given that training data $\neq$ test data?

Challenges in statistical learning and probabilistic solutions

Probabilistic tools and viewpoints offer a formal way to manage and quantify uncertainty

In particular,

- **Issue/Need:** Training data $\neq$ test data
  - *Solution:* Assume training and test data are *randomly drawn* from same distribution
- **Issue/Need:** Uncertainty in prediction
  - *Solution:* Model Y as a *random variable*, and predict Y conditioned on X
- **Issue/Need:** Choosing among many models
  - *Solution:* Identify the "most likely" model given data, or update our belief about the "best model" based on observed data

We will discuss these aspects in more detail throughout the course

Probability in everyday examples

- Coin toss
  - Possible outcomes are Head or Tail; each has probability 0.5
- Die roll
  - Possible outcomes $\{1, 2, \ldots, 6\}$; each has probability $1/6$
- Y chromosomes in the US childbirths[1]
  - About 51.2% of births are to babies with Y chromosomes, and 48.8% do not
  - The probability of having a baby with a Y chromosome is 0.512
- Commute time
  - Commute may take 30 minutes on average, but can vary with traffic, weather, etc.
- Subjective probabilistic beliefs
  - You may personally estimate the likelihood of a stock price rising or falling, based on your own analysis or expert opinions
  - This kind of probability reflects beliefs rather than strict long-run frequencies

[1] Source: CDC National Vital Statistics Reports, Births: Final Data for 2023

Formalizing probability: Sample space and events

- **Outcome:** a possible result of an experiment or trial
- **Sample space**: the set of all possible outcomes, often denoted by Ω
  - e.g., $\{H, T\}$, $\{1, 2, 3, 4, 5, 6\}$
- **Event:** a subset of Ω
  - e.g., for $\Omega = \{H, T\}$: $\emptyset$, $\{H\}$, $\{T\}$, $\{H, T\}$
  - e.g., for $\Omega = \{1, 2, 3, 4, 5, 6\}$: $\{6\}$, $\{1, 2\}$, $\{2, 4, 6\}$, ...
- **Probability**[2]**:** a map P that assigns a number in $[0, 1]$ to each event such that
  - $P(\Omega) = 1$;
  - For disjoint events $A_1, A_2, \ldots,$ $P\left(\bigcup_i A_i\right) = \sum_i P(A_i)$
    - Simply put, if $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$

[2] A formal, mathematically rigorous definition of probability measure is beyond the scope of STA 35C

Where we are headed

- **Throughout the course**:
  - Learn key ideas of regression, classification, clustering, and more
  - Practice implementing methods and interpreting results
  - Assess when each method is or is not appropriate
- **Immediate next steps (Week 1)**:
  - Refresh core probability concepts (from STA 35A/B)
  - Deepen understanding of conditional probability and explore Bayesian ideas
- **Before next lecture**:
  - Complete "Homework 0" at your earliest convenience and seek help if needed
  - Ensure you have access to R and RStudio